

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Scenario-Based

Code of Conduct:

*I Y.GEETHA REDDY(192572053)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:Geetha

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

ANS.

1.

Department	Subnet	Network Address	Broadcast Address
School of Computing	/25	192.168.100.0	192.168.100.127
School of Electronics	/26	192.168.100.128	192.168.100.191
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.223

2.

Department	Valid Host Range
School of Computing	192.168.100.1 – 192.168.100.126
School of Electronics	192.168.100.129 – 192.168.100.190
School of Mechanical Engineering	192.168.100.193 – 192.168.100.222

3.

Department	Usable Hosts
School of Computing	126
School of Electronics	62
School of Mechanical Engineering	30

4.Unused IP Range: 192.168.100.224 – 192.168.100.255

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

ANS.

1.

Department	Network Address	Broadcast Address
Software Development (/24)	10.10.0.0	10.10.0.255
Quality Assurance (/25)	10.10.1.0	10.10.1.127
Human Resources (/26)	10.10.1.128	10.10.1.191
Executive Management (/27)	10.10.1.192	10.10.1.223

2.

Department	Host Range
Software Development	10.10.0.1 – 10.10.0.254
Quality Assurance	10.10.1.1 – 10.10.1.126
Human Resources	10.10.1.129 – 10.10.1.190
Executive Management	10.10.1.193 – 10.10.1.222

3.

Department	Usable Hosts
Software Development	254
Quality Assurance	126
Human Resources	62
Executive Management	30

4. Unused IP Range: 10.10.1.224 – 10.10.255.255

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

ANS.

1. 2001:db8:abcd::/64

2. 2001:0db8:abcd:0000:0000:0000:0000:0000

3. 2001:db8:abcd:0000::/64

4. Host bits = 64

Number of Hosts = 2^{64}

= 18,446,744,073,709,551,616

5. 18,446,744,073,709,551,616 addresses

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries: 192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

ANS.

1. Destination IP = 192.168.2.45

2. 192.168.2.0/24

3. Router forwards the packet through the interface connected to the 192.168.2.0/24 network.

4. 0.0.0.0/0

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.

4. Identify the **default gateway** that hosts in each LAN should use for communication.

ANS.

1.Administration LAN (192.168.10.0/26)

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63

Finance LAN (192.168.10.64/26)

- **Network Address:** 192.168.10.64
- **Broadcast Address:** 192.168.10.127

2.Administration LAN

192.168.10.1 – 192.168.10.62

Finance LAN

192.168.10.65 – 192.168.10.126

3.Administration LAN

- PC1 → 192.168.10.2
- PC2 → 192.168.10.3
- PC3 → 192.168.10.4

Finance LAN

- PC1 → 192.168.10.66
- PC2 → 192.168.10.67
- PC3 → 192.168.10.68
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4.Administration LAN

Default Gateway = 192.168.10.1

Finance LAN

Default Gateway = 192.168.10.65

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided

Clarity of Response	15%	Response is clear, structured, and logically presented	General ly clear with minor issues	Somewh at unclear or poorly structure d	Disorganiz e d and difficult to understand
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